

Subject Applied econometrics

For obtaining a Master of Excellence

« Economie Appliquée et Data Science »

Practical exercises on R :

The Gravity Model in international Trade

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Note:

Since no single database provided all the information required for this study, the dataset was constructed by combining data from several sources, with the World Bank database serving as the primary source. The collected data were subsequently verified, organized, and harmonized in Microsoft Excel before being converted into CSV format for econometric processing in R.

The present report is accompanied by the database developed for this research as well as the R script containing all estimation procedures, including the Pooled OLS, Between, Fixed Effects, and Random Effects models, in addition to the econometric tests used for model selection.

Objective

The objective of this mini-project is to analyze the determinants of international trade between six countries: Morocco, Egypt, France, Spain, Saudi Arabia, and China during the period 2010–2020.

To achieve this objective, the Gravity Model of International Trade is used to examine the impact of the Gross Domestic Product (GDP) of the trading partners and the geographical distance between them on the volume of bilateral trade.

The main hypothesis is that countries with larger economies trade more with each other, while greater geographical distance reduces trade flows.

Methods (Econometric Explanation)

This study uses panel data, which combines both a cross-sectional dimension (country pairs) and a time dimension (2010–2020).

The estimated gravity model is:

$$\ln(\text{Trade}_{ij,t}) = \beta_0 + \beta_1 \ln(\text{GDPA}_{ij,t}) + \beta_2 \ln(\text{GDPB}_{ij,t}) + \beta_3 \ln(\text{Distance}_{ij}) + u_{ij,t}$$

Where:

- **Trade** = bilateral trade volume between countries i and j ;
- **GDPA** = GDP of country A;
- **GDPB** = GDP of country B;
- **Distance** = geographical distance between the two countries;
- **u** = error term.

Four panel data estimators were applied:

1. **Pooled OLS Model**
2. **Between Estimator**
3. **Within Estimator (Fixed Effects Model)**

4. Random Effects Model

To select the most appropriate model, three econometric tests were performed:

- **F-test** (Fixed Effects vs. Pooled OLS)
- **Lagrange Multiplier (LM) Test** (Random Effects vs. Pooled OLS)
- **Hausman Test** (Fixed Effects vs. Random Effects)

Results (Statistical Interpretation)

The Pooled OLS estimation shows positive coefficients for **GDPA** ($\beta = 0.563$) and **GDPB** ($\beta = 0.569$), while the coefficient of **Distance** ($\beta = -0.498$) is negative. All p-values are below 0.05, indicating that the explanatory variables are statistically significant.

The coefficient of determination ($R^2 = 0.844$) indicates that approximately **84.4%** of the variation in trade flows is explained by GDP and distance.

The Between estimator also provides positive coefficients for GDP variables and a negative coefficient for distance. The model achieves an **R^2 of 87.5%**, suggesting a strong explanatory power.

The Within (Fixed Effects) model produces positive coefficients for both GDP variables. The distance variable is excluded because it remains constant over time for each country pair.

The econometric tests produced the following results:

- F-test: significant (p-value < 0.05)
- LM Test: significant (p-value < 0.05)
- Hausman Test: significant (p-value = 0.000072 < 0.05)

Therefore, the Fixed Effects (Within) model is selected as the most appropriate specification for this study.

Economic Interpretation

The results are consistent with the predictions of the Gravity Model of International Trade.

The positive coefficients of **GDP_A** and **GDP_B** indicate that larger economies tend to trade more with each other. Economic growth increases production capacity, consumption, and demand for imported goods, thereby stimulating international trade.

The negative coefficient of **Distance** confirms that geographical distance acts as a barrier to trade. Greater distances generally imply higher transportation costs, longer delivery times, and increased logistical constraints, which reduce trade flows between countries.

Finally, the Hausman test suggests that country-pair specific characteristics significantly influence trade flows. As a result, the Fixed Effects model is the most suitable framework for analyzing trade relationships among Morocco, Egypt, France, Spain, Saudi Arabia, and China during the 2010–2020 period.